Let R be a Commutative Ring with $1 \neq 0$ in this paper.

Def. In R, define a set

$$\eta(R) \overset{def}{=} \{x \in R \mid x \text{ is nilpotent}\}. \quad (\text{i.e., } x^m = 0 \text{ for some } m \in \mathbb{N})$$

Actually $\eta(R)$ is an ideal.

Given $a, b \in \eta(R)$, $a^m = b^n = 0$ for some $m, n \in \mathbb{N}$, so,

$$(a+b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} \cdot a^{m+n-k} \cdot b^k$$

If $0 \leq k < n$, then $m+n-k > m$, so $a^{m+n-k} = 0$

If $k \geq n$, then $b^k = 0$. Therefore, $(a+b)^{m+n} = 0$, or $a+b \in \eta(R)$.

And, $(-a)^m = (-1)^m \cdot a^m = 0$, so $-a \in \eta(R)$.

Also, since $(ab)^{mn} = a^{mn} \cdot b^{mn} = 0$ by the Commutativity, $ab \in \eta(R)$.] Subring.

Given $r \in R$ and $a \in \eta(R)$ with $a^m = 0$, $(ra)^m = r^m a^m = r^m \cdot 0 = 0$, so $ra \in \eta(R)$.] Ideal.

Lemma. $R/\eta(R)$ has no nilpotent element except zero.

Proof. Let $a + \eta(R) \in R/\eta(R)$ be given. If $(a+\eta(R))^n = 0$ for some $n \in \mathbb{N}$, then

$$(a+\eta(R))^n = a^n + \eta(R) = 0 + \eta(R) \iff a^n \in \eta(R)$$

Then, a^n is a nilpotent element, so a is nilpotent, or $a \in \eta(R)$.

Thus, the only nilpotent element is $0 + \eta(R)$ in $R/\eta(R)$.

Prop. Let R be a Commutative Ring with $1 \neq 0$. Given $P(x) = a_n x^n + \cdots + a_1 x + a_0 \in R[x]$,

$P(x)$ is a unit in $R[x]$ iff a_0 is unit and $a_1, \cdots, a_n$ are nilpotent.

$P(x)$ is a nilpotent in $R[x]$ iff $a_0, \cdots, a_n$ are nilpotent.

Proof. Since sum of nilpotent, if $a_1, \cdots, a_n$ are nilpotent, then $n = a_1 x + \cdots + a_n x^n$ is nilpotent.

So, $a_0 + n$ is a unit, being sum of unit and nilpotent.

Conversely, if $P(x)$ is a unit, then for some $g(x) = b_m x^m + \cdots + b_0 \in R[x]$, $P(x) \cdot g(x) = 1$.

So, $a_n b_m x^{n+m} = 0$,

「Thm. Let R be a Commut. Ring with $1 \neq 0$.

Then, $\eta(R) \subseteq \bigcap_{\substack{P \subset R \\ P \text{ is Prime ideal.}}} P$. (Actually, these are Same).

Proof. Let P be a Prime Ideal. Suppose that for some non-zero nilpotent $a \in R$, $a \notin P$.
That is, $a+P \neq P$ in R/P. Since a is nilpotent, $m \in \mathbb{N}$ be the smallest number s.t. $a^m = 0$. Now,

$$P = a^m + P = (a+P)\cdot(a^{m-1} + P),$$

by the minimality of m, $a^{m-1} + P \neq P$, so $a+P$ is zero divisor.
But, since P is prime ideal, R/P is integral domain. So Contradiction.

Another proof: $0 = a^m = a \cdot a^{m-1} \in P$, so a or $a^{m-1} \in P$.
$a \in P$, then Clear. $a^{m-1} \in P$, then a or $a^{m-2} \in P$.
Inductively, $a \in P$.

「Prop. If $a \in R$ is nilpotent, then $1 - ab$ is a unit, for any $b \in R$
Proof. Let $m \in \mathbb{N}$ s.t. $a^m = 0$. Then,

$$(1-ab)\cdot(1 + ab - (ab)^2 + \cdots - (-ab)^{m-1}) = 1 + (-1)^m \cdot (ab)^m = 1 + 0 = 1.$$

Def. Let $I \subseteq R$ be an Ideal. Define the radical of I

$$\operatorname{rad} I \overset{\text{def}}{=} \{ r \in R \mid r^n \in I \text{ for some } n \in \mathbb{N} \}.$$

Then, $\operatorname{rad} I$ is an Ideal of R satisfying

$$(\operatorname{rad} I)/I = \eta(R/I).$$

And, I is called a radical Ideal if $\operatorname{rad} I = I$.

Proof. 1). $\operatorname{rad} I$ is an Ideal.

Given $a, b \in \operatorname{rad} I$, $a^m, b^n \in I$ for some $m, n \in \mathbb{N}$. So,

$$(a+b)^{m+n} = \sum_{k=0}^{m+n} \binom{m+n}{k} a^{m+n-k} \cdot b^k \in I,$$

because since I is ideal, if $0 \le k < n$, $\binom{m+n}{k} \cdot a^{m+n-k} \cdot b^k = \binom{m+n}{k} \cdot b^k \cdot a^{n-k} \cdot a^m \in I$, and if $k \ge n$, $\binom{m+n}{k} a^{m+n-k} \cdot b^{k-n} \cdot b^n \in I$. So, the sum is contained in I.

And, similarly, $(ab)^{m+n} = a^{m+n} \cdot b^{m+n} \in I$. $\lrcorner$ Subring.

And, given $r \in R$ and $a \in \operatorname{rad} I$,

$$(ra)^m = r^m a^m \in I \quad \lrcorner \text{ Ideal.}$$

Clearly, $I \subseteq \operatorname{rad} I$. Define a map

$$\varphi : \operatorname{rad} I \to \eta(R/I) : a \mapsto a + I$$

Then, φ is well-defined onto Homomorphism, because given $a \in \operatorname{rad} I$, $a^m \in I$ for some $m \in \mathbb{N}$, so $(a+I)^m = a^m + I = I$, or $a + I \in \eta(R/I)$. So well-defined. Homomorphism is clear, and onto is clear. Moreover,

$$\varphi(a) = 0 \iff a + I = I \iff a \in I, \text{ so } \ker \varphi = I.$$

Finally, the first-Isomorphism gives $\operatorname{rad} I / I \cong \eta(R/I)$.

Actually,
$$\bar{\varphi} : \operatorname{rad} I / I \to \eta(R/I) : a + I \mapsto a + I$$

is identifying, $\operatorname{rad} I / I = \eta(R/I)$.

Nilpotent with the Polynomial Ring.

Lemma. Let R be a Ring with identity, and $a, b \in R$.
If a is nilpotent and b is unit and a, b commute,
then $a+b$ is unit.

Proof. Choose $n \in \mathbb{N}$ s.t. $a^n = 0$. Then, since $a = abb^{-1} = bab^{-1} \Rightarrow b^{-1}a = ab^{-1}$,
and $b^n - a^n = (b+a) \cdot \left(\sum_{k=0}^{n-1} (-1)^k \cdot b^{n-k-1} \cdot a^k \right) = b^n$, therefore

$$(a+b)^{-1} = \left(\sum_{k=0}^{n-1} (-1)^k \cdot b^{n-k-1} \cdot a^k \right) \cdot b^{-n}.$$